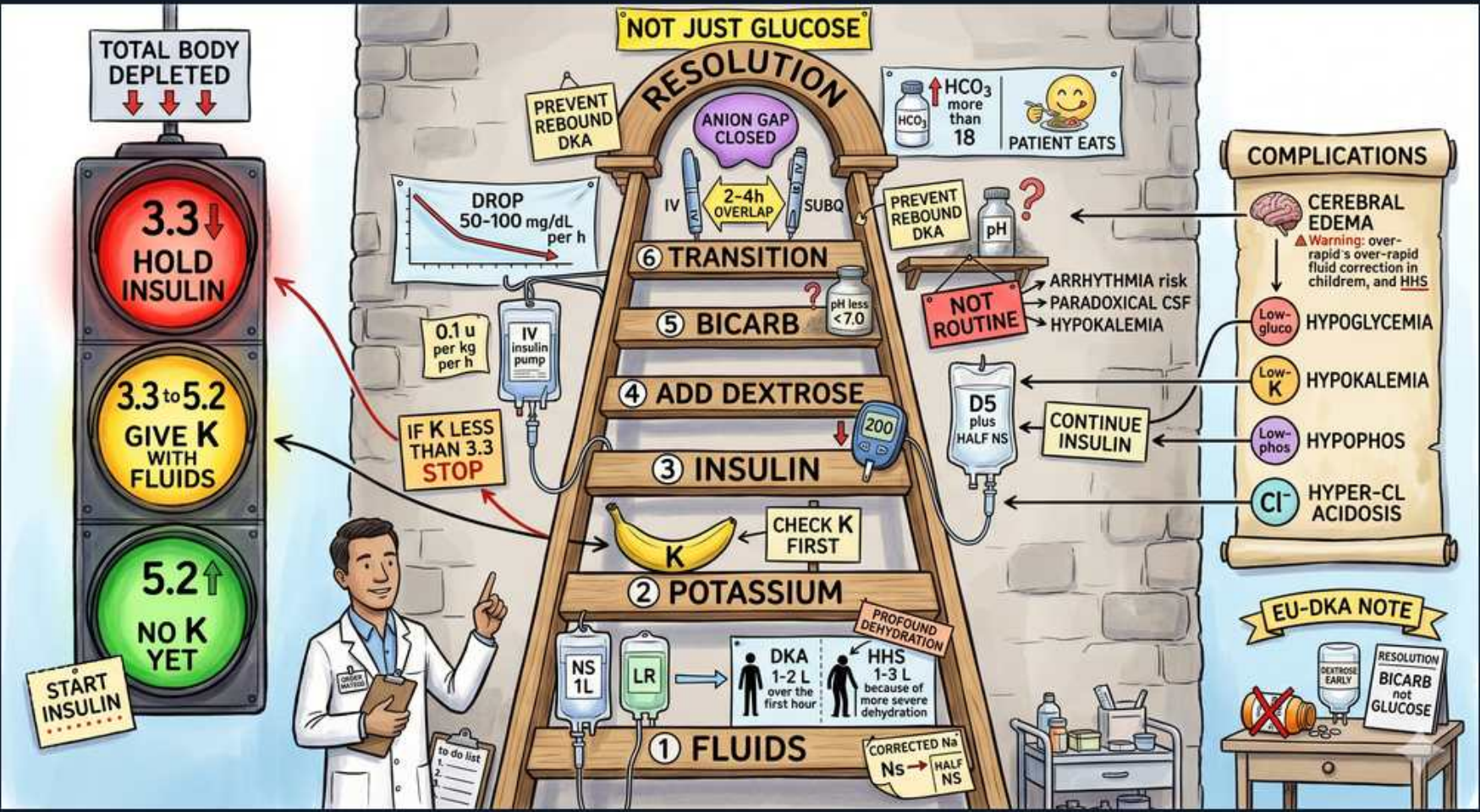


DKA / HHS Acute Management — The Treatment Ladder



1. Step 1 — Fluids first

WHAT YOU SEE

Rung 1 FLUIDS, NS bags, DKA 1-2L / HHS 3-1L

WHAT IT ENCODES

Start IV isotonic fluids before anything else. DKA needs ~1-2 L early; HHS patients are more profoundly volume depleted (often >8-10 L total). Correct sodium for glucose.

BOARD TRAP

Giving insulin before fluids in a hypotensive patient worsens shock — fluids and potassium precede insulin.

2. Step 2 — Potassium

WHAT YOU SEE

Rung 2 POTASSIUM, banana, 'check K first'

WHAT IT ENCODES

Check potassium before insulin. Insulin drives K into cells and can cause lethal hypokalemia. Total body K is depleted despite normal/high serum value.

BOARD TRAP

A normal serum K in DKA actually signals significant total-body deficit — do not be reassured.

3. Total body depleted

WHAT YOU SEE

TOTAL BODY DEPLETED scale icon

WHAT IT ENCODES

Both K and water are total-body depleted in DKA/HHS even when labs look normal high, because acidosis and osmotic diuresis shift and waste them.

BOARD TRAP

Treating only the serum number, not the deficit, leads to crash hypokalemia after insulin starts.

4. K traffic light — hold <3.3

WHAT YOU SEE

Red light 3.3, HOLD INSULIN, give K

WHAT IT ENCODES

If K <3.3 mEq/L: HOLD insulin and replace potassium first (10-20 mEq/hr) until K rises. Starting insulin here can cause fatal arrhythmia.

BOARD TRAP

Most-missed step — insulin started at K<3.3 is the classic fatal error in DKA boards.

5. K traffic light — 3.3–5.2

WHAT YOU SEE

Yellow light, give K with fluids

WHAT IT ENCODES

K between 3.3 and 5.2: give potassium (20-30 mEq) in each liter of fluid AND start insulin together.

BOARD TRAP

Forgetting to add K in the 3.3-5.2 window leads to hypokalemia once insulin shifts K intracellularly.

6. K traffic light — >5.2

WHAT YOU SEE

Green light 5.2, no K yet

WHAT IT ENCODES

K >5.2: do not give potassium yet; proceed with insulin and recheck K every 2 hours.

BOARD TRAP

Adding K when >5.2 risks hyperkalemia — withhold until it falls into range.

7. Step 3 — Insulin

WHAT YOU SEE

Rung 3 INSULIN, 0.1 u/kg/hr drip

WHAT IT ENCODES

IV regular insulin infusion ~0.1 u/kg/hr (a bolus is optional). Target glucose fall ~50-100 mg/dL/hr. Continue insulin until the anion gap closes, not just until glucose normalizes.

BOARD TRAP

Stopping insulin when glucose normalizes but gap still open allows DKA to recur — keep insulin going.

8. Step 4 — Add dextrose

WHAT YOU SEE

Rung 4 ADD DEXTROSE, D5 at glucose ~200

WHAT IT ENCODES

When glucose reaches ~200 mg/dL, switch to D5-containing fluids and continue insulin to keep clearing ketones/closing the gap.

BOARD TRAP

Letting glucose drop without adding dextrose causes hypoglycemia while ketoacidosis is still resolving.

9. Step 5 — Bicarbonate

WHAT YOU SEE

Rung 5 BICARB, 'not routine', pH<6.9

WHAT IT ENCODES

Bicarbonate is NOT routine. Consider only if pH <6.9. Otherwise insulin and fluids correct acidosis on their own.

BOARD TRAP

Routine bicarb risks paradoxical CSF acidosis, hypokalemia, and arrhythmia — reserved for pH<6.9.

10. Step 6 — Transition

WHAT YOU SEE

Rung 6 TRANSITION, overlap SubQ/IV

WHAT IT ENCODES

Transition to subcutaneous insulin only after the anion gap is closed and the patient can eat; overlap SubQ and IV by ~2 hours to prevent rebound.

BOARD TRAP

Stopping the drip before SubQ takes effect (no 1-2 h overlap) causes rebound DKA.

11. Resolution criteria

WHAT YOU SEE

RESOLUTION banner, gap closed, patient eats

WHAT IT ENCODES

DKA resolved when anion gap is closed, bicarbonate >18, and the patient can eat. Glucose alone does not define resolution.

BOARD TRAP

Normal glucose with a still-open gap is NOT resolution — keep treating.

12. Cerebral edema

WHAT YOU SEE

COMPLICATIONS — brain, over-rapid correction

WHAT IT ENCODES

Cerebral edema follows over-rapid correction of glucose/osmolality, mostly in children. Limit glucose fall to 50-100 mg/dL/hr.

BOARD TRAP

Headache/altered mentation hours into therapy = cerebral edema, the feared pediatric DKA complication.

13. Other complications

WHAT YOU SEE

Low K, low phos, hyper-Cl acidosis tags

WHAT IT ENCODES

Watch hypokalemia, hypophosphatemia, hypoglycemia, and a non-gap hyperchloremic acidosis from large saline volumes during recovery.

BOARD TRAP

A persistent non-anion-gap acidosis after gap closure is dilutional hyperchloremic, not ongoing DKA.

14. Euglycemic DKA note

WHAT YOU SEE

EU-DKA NOTE box, bicarb + glucose

WHAT IT ENCODES

Euglycemic DKA (SGLT2i, pregnancy, starvation) presents with gap acidosis and near-normal glucose — still needs insulin plus dextrose from the start.

BOARD TRAP

Withholding insulin because glucose is normal misses euglycemic DKA — treat the gap.
